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# Decision Tree for Spam Detection:

import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.feature_extraction.text import CountVectorizer

df = pd.read_csv("email.csv")

x = df["Message"]
y = df["Category"]

le = LabelEncoder()
y = le.fit_transform(y)

cv = CountVectorizer()
x = cv.fit_transform(x)

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2)

model = DecisionTreeClassifier(criterion="gini", max_depth=3)
model.fit(x_train, y_train)

y_pred = model.predict(x_test)
print(y_pred)
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# Random Forest for Spam Detection:

import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.feature_extraction.text import CountVectorizer

df = pd.read_csv("email.csv")

x = df["Message"]
y = df["Category"]

le = LabelEncoder()
y = le.fit_transform(y)

cv = CountVectorizer()
x = cv.fit_transform(x)

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2)

model = RandomForestClassifier(criterion="gini", n_estimators=10)
model.fit(x_train, y_train)

y_pred = model.predict(x_test)
print(y_pred)
[0 0 0 ... 0 1 0]
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